

Spiroscopic Recursive Transformers

Structured divergence, cyclic function orbits, and an honest experimental program

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Abstract

Looped neural models are usually designed to converge: repeated application of a tied map is useful when the local dynamics approach a stable solution. Spiroscopic Recursive Transformers (SRTs) invert that default. They schedule bounded expansion, rotation, and selection before a contractive readout. The intended object is not a model that merely grows its activations, but a controlled traversal of a function or coordinate orbit: a compact, deterministic proposal mechanism whose visited states can be pooled, ranked, or handed to a conventional calibrator.

This note records the SRT proposal, its relation to convergent recurrent models, and the experimental discipline developed around it. The central empirical conclusion is deliberately limited. The work produced useful diagnostics, counterfactuals, and benchmark-design constraints, but has not established a broad SRT capability advantage. A feedback-navigator experiment reached a fresh-root effect of +0.0918 against a preregistered +0.10 threshold and is therefore non-confirmatory. Other variants either lost to simple baselines or exposed an invalid task mechanism before training. Sparse positive exploratory signals remain hypotheses, not results. The main contribution is a falsifiable research program for testing when phase-structured exploration supplies information that a scan-and-pool baseline cannot recover.

1. Why schedule divergence?

Tiny Recursive Models (TRMs), Hierarchical Recursive Models (HRMs), deep equilibrium systems, and related tied-loop architectures normally derive their interpretation from contraction. If a recurrent update F has a Jacobian with spectral radius below one near an answer, repeated application can be read as calibration toward a fixed point. The same stability that makes a tied gain interpretable also limits the loop's role as a proposal generator.

SRTs add a divergent primitive to this vocabulary. In a loop schedule, C denotes a convergent or consolidating phase and D denotes a bounded divergent phase. $D^k C^m$ reads as "scatter for k steps, then collapse for m ." This is not a claim that arbitrary divergence is useful. It is a claim that the schedule, collar, and selection rule should be explicit architectural variables rather than accidental by-products of an unstable residual stream.

The useful contrast is between unstructured norm growth and a controlled orbit. Pure radial growth explores almost nothing: it is a single drip thrown away from the canvas. A complex eigenvalue pair with modulus above one gives radial expansion plus angular motion. The phase component is what visits different relative coordinate alignments. The contraction phase, a viability collar, and a downstream selector make those visits testable rather than decorative.

2. A minimal SRT state model

Let a latent state z be decomposed relative to an anchor a into a radius r and direction θ :

$$z - a = r * u(\theta), \quad ||u(\theta)|| = 1.$$

An SRT update separates a phase-bearing lane from a conventional content lane:

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theta_(t+1) = theta_t + omega_t(x, h_t) + epsilon_t
r_(t+1)     = collar(r_t + delta_t)
h_(t+1)     = G(h_t, u(theta_(t+1)), x)

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Here ω may be exogenous, learned, or content-modulated; collar clips, reflects, or otherwise confines radius to a registered viable interval; and G may be an ordinary transformer block, MLP, or recurrent calibrator. The phase lane avoids asking LayerNorm to preserve information encoded solely as residual-stream magnitude. In the most tractable version, r is bookkeeping and only the directional state is injected into the network proper.

The term "spiroscopic" is intentional. The model need not draw a literal spirograph in activation space. What matters is a structured sequence of relative alignments among coordinates, injections, or low-rank function components. A fixed gear train can produce many views of one function without claiming to search an unconstrained weight space.

3. From activation orbits to function orbits

The first SRT implementations explored input and coordinate phase. A stronger extension holds a base map fixed and visits a low-rank function manifold:

$$A(q) = A_0 + \beta \sum_k q_k L_k R_k.$$

The coefficients q follow a scheduled orbit generated by overlapping rotations. Each visit materializes a distinct, collared transformation while sharing A_0 and six seeded rank-2 components. A fixed-width feature recipe can then be evaluated as

$$[x, \sin(A(q)x), \cos(A(q)x), \tanh(A(q)x)].$$

This changes the claim precisely. The loop explores parameterized views of a shared function family, not arbitrary parameter combinations. With selection or pooling over visits, the mechanism becomes a compact structured ensemble. Without a task on which the identity or order of a parameterization matters, apparent gains can reduce to ordinary averaging or variance reduction.

SpiroLift is the practical, tabular version of this idea. It visits a fixed budget of 24 candidates, evaluates one deterministic ridge classifier per candidate on a calibration split, and locks the first feature map that clears preregistered constraints on macro-F1, worst-class recall, expected calibration error, and bootstrap preservation. Otherwise it returns raw features. The application is an exploratory feature foundry; only sealed Python runs can support research claims.

4. The canvas: collars, collapse, and selection

An SRT has three safeguards.

First, a collapse mechanism. Expansion without selection is entropy. The architecture must either pool useful visits, choose a viable visit, or contract to a readout state. This resembles a deterministic analogue of expansion-then-denoising, but does not imply that deterministic chaotic itinerancy is superior to stochastic noise.

Second, a confinement width. Pollock's visual metaphor is useful only because the drips have an edge. A collar defines that edge in latent or function space. It is a hyperparameter and a pretraining constraint, not merely a statistic measured after an unstable run.

Third, an orbit diagnostic. Two-dimensional trajectory projections admit box-counting and coverage statistics. A point-like contraction and indiscriminate plane filling are both poor targets. The speculative regime is a bounded, structured middle. Such geometry is a diagnostic only; it cannot substitute for task performance or a counterfactual control.

5. What the experiments did and did not show

The experimental program used a sequence of atlas, Rotor-Lock, Curve-Lock, and feedback-navigator probes. Their purpose was to distinguish four explanations: a genuine order-sensitive orbit computation; cyclic scan-and-pool; standard ensemble averaging; and a benchmark whose label leaks through simpler statistics.

The early torque atlas delivered the most important negative lesson. Circular pooling transferred only to origin-invariant rotational targets. Temporal-origin accuracy remained pinned at 66.67 percent across torque policies. The proper conclusion was "cyclic scan-and-pool, not rotational intelligence," and no additional torque complexity was justified without labels generated by content-dependent angular dynamics.

A later learned static atlas did not beat a raw trace baseline. A feedback-navigator variant produced a promising discovery signal, but the sealed fresh-root confirmation effect was +0.0918 against a frozen requirement of +0.10. It is close numerically but fails the gate and is reported as non-confirmatory. A subsequent diagnostic objective exhibited label collapse before training; that task was closed rather than interpreted as a model outcome. An early 33.3 percent result was also treated as suspect rather than promoted.

Accordingly, there is no validated evidence here that SRTs broadly improve transformers, TRMs, LDTs, tabular classifiers, storyworld play, or general search. There is evidence that the controls are capable of finding seductive false positives and that SRT proposals should remain coupled to a conventional TRM or LDT calibration stage until a task-specific mechanism passes confirmation.

6. A sharper target: cyclic functional holonomy

One promising benchmark family is a path-ordered product of noncommuting transforms. Given an ordered sequence $A_{\pi(1)}, \dots, A_{\pi(T)}$, define

$$P = A_{\pi(T)} \dots A_{\pi(1)}, \quad y = Q(\text{trace}(P), \text{trace}(P^2), \text{trace}(P^3)).$$

The trace-class label is important. A vector-valued composition label $A_{\pi(T)} \dots A_{\pi(1)}x$ is generally not invariant to cyclic shifts of the order. Trace, spectrum, determinant, and related conjugation-invariant functionals are cyclic. The proposed label is therefore analogous to a Wilson-loop observable: it retains order sensitivity in a non-abelian product while respecting cyclic re-rooting of a closed path.

That analogy creates obligations, not marketing. The manifest should require $T \geq 3$, distinct operators per orbit or explicit repeat handling, non-palindromic-up-to-rotation orderings, a noncommutativity floor based on pairwise commutators, and scale normalization so product magnitude cannot carry the label. Joint label quantization must account for correlations among trace power sums. A reversal control must exclude near-normal transform families where it becomes vacuous.

Under black-box matrix-vector access, a direct baseline should use a registered probe estimator for trace. Its ceiling must be measured before model training. A same-set order shuffle is the key counterfactual: it preserves every candidate transformation and changes only its ordering. No claimed SRT effect survives unless candidate-set hashes match exactly, order shuffling causes the preregistered degradation, and matched non-orbit baselines fail the same target.

7. Training signal and hybrid use

SRT is most plausible as a proposal stage, not as a replacement for calibration. A hybrid has three responsibilities:

1. SRT generates a bounded set of phase or function-orbit states.

2. A TRM calibrates or verifies candidate consequences.

3. An LDT stores the selected consequence, score, and rejection reason as an auditable trace.

The training signal should reward conditional value of a visit, not visual spiral quality. Candidate objectives include improvement over a matched static or shuffled orbit, prediction of which visit will lock, orbit-order sensitivity on synthetic holonomy tasks, calibration of abstention when no visit is viable, and diversity conditional on usefulness. Every such objective needs a null lane: label shuffle, duplicated rows, linear tasks, scale inflation, and no-lock fallback.

For storyworlds, the interface should be an estimated-effects packet rather than hidden free-form speculation. Each action option produces a bounded vector of predicted changes to named variables, uncertainty, and an orbit identifier. The LLM or TRM remains the decision-maker; the SRT proposes structured counterfactual effect patterns. Success would require better calibrated downstream choices, not more colorful trajectories.

8. Registered next tests

The next confirmatory study should pre-register one synthetic mechanism lane and a descriptive external-validity lane. The synthetic lane should use four hidden function classes by four first-hit bins, a balanced 16-way joint label, and a later higher-scoring viable decoy from another class. Discovery roots, confirmation roots, holdout roots, candidate budget, collar, and selection rule must be sealed in advance.

Support should require all of the following: ordered SRT macro-F1 at least +0.10 over both random and Sobol coefficient controls with a positive paired root-block 95 percent interval; an original-label loss of at least 0.20 under the same-set shuffle; shuffled-order counterfactual macro-F1 of at least 0.75; exact candidate-set preservation; and a positive holdout mean on at least three of four roots. An evaluation review must invalidate the run before outcomes are interpreted if invariants, nulls, or data provenance fail.

Real datasets such as Breast Cancer, Wine, and Digits are useful descriptive checks for feature locking, runtime, worst-class recall, calibration, and selection stability. They cannot rescue a failed synthetic mechanism claim.

9. Conclusion

The SRT idea is worth keeping because it is now narrow enough to fail cleanly. Its distinctive proposal is not "divergence helps" or "spirals are intelligent." It is that a bounded, phase-structured orbit can expose order-dependent alternatives that a calibrating loop can evaluate, provided the label generator makes that order dependence necessary and controls rule out pooling and leakage.

The present record is mixed and predominantly negative. That is useful. It rules out several attractive but weak stories, preserves the sparse exploratory observations without overstating them, and leaves a concrete test: demonstrate an order-sensitive gain on a sealed noncommutative functional task, or retire the mechanism.

References

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